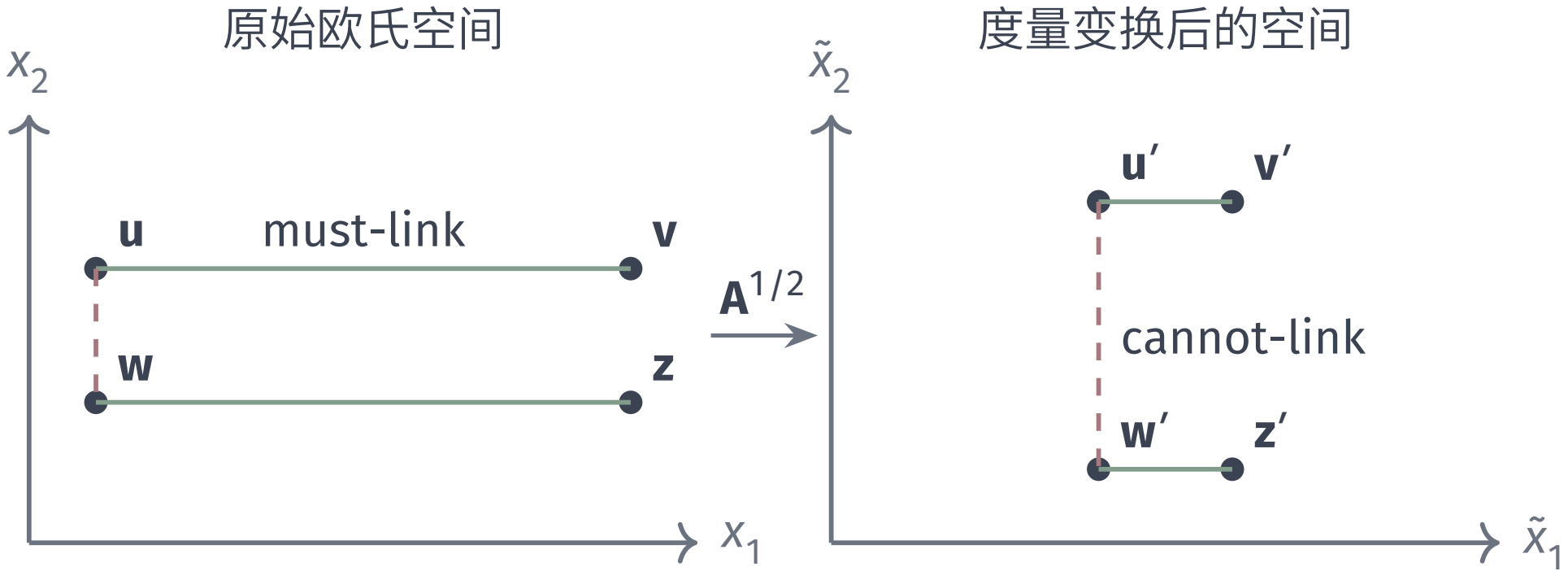

$\|\mathbf{u}-\mathbf{v}\|_2^2 = 16, \quad \|\mathbf{u}-\mathbf{w}\|_2^2 = 1$

$d_{\mathbf{A}}^2(\mathbf{u},\mathbf{v}) = 1, \quad d_{\mathbf{A}}^2(\mathbf{u},\mathbf{w}) = 4$